
Multimodal Anomaly Detection using Vision Language Models and Multimodal Large Language Models

Literature Survey & Proposed Research Direction

Prakash Kumar Sarangi | M251250CS

Guide: Dr. Pranesh Das

Department of Computer Science & Engineering
National Institute of Technology Calicut

Outline

1. Problem & Motivation
2. Background — CLIP, MLLMs, MVTec AD
3. Literature Review — 5 Key Papers
4. Comparative Analysis
5. Identified Research Gaps
6. Proposed Research Direction
7. Expected Contributions & Timeline

Problem & Motivation

Current Limitations

- Single-modality (image-only) approaches
- Require training data per product category
- No explanation of detected defects
- Human inspection is slow & error-prone

The Opportunity

- Vision-Language Models enable zero-shot detection
- MLLMs can explain anomalies in natural language
- No training data needed — just text prompts
- Interpretable, scalable quality inspection

Research Question

How can VLMs and MLLMs be leveraged for zero-shot anomaly detection in industrial quality inspection — achieving robust detection AND interpretable explanations without task-specific training?

Background

CLIP (ICML 2021)

- 400M image-text pairs
- Contrastive learning
- Image + Text encoders
- Shared embedding space
- Zero-shot via cosine similarity

MLLMs (LLaVA)

- Vision encoder + LLM
- See images & answer questions
- Natural language reasoning
- Open-source (NeurIPS'23)
- Describe, explain, reason

MVTec AD Benchmark

- Standard benchmark (CVPR 2019)
- 15 industrial categories
- 5,354 high-res images
- Pixel-level annotations
- Metric: AUROC (0-100%)

Key Metric: AUROC — Area Under ROC Curve. Higher = better. 99.1% (PatchCore) is ceiling with training.

PatchCore — Towards Total Recall in Industrial AD

Roth et al. | CVPR 2022 | Traditional Baseline

Method

1. Extract patch features from pre-trained CNN
2. Store in memory bank ("normal" dictionary)
3. Coreset subsampling — keep 1% representative features
4. Test: KNN distance to bank = anomaly score
5. Patches far from all normals = anomalous

Results

99.1% Image AUROC on MVTec AD
Strongest traditional baseline

Limitations

- X Needs normal training images per category
- X Cannot generalize to unseen products
- X No natural language explanations

Takeaway: Upper-bound baseline — can we approach 99% without any training?

WinCLIP — Zero-/Few-Shot Anomaly Classification & Segmentation

Jeong et al. | CVPR 2023 | CLIP-Based Zero-Shot

Method — Two Key Innovations

Use CLIP's ability to compare images and text

1. Compositional Prompt Ensemble (CPE):

Templates x State words -> ~20 prompts -> averaged prototypes

e.g., "a photo of a [damaged/flawless] [screw]"

2. Multi-Scale Sliding Window:

Image -> overlapping windows (small + mid + global)

Each window -> CLIP features -> compare with text prototypes

-> Pixel-level anomaly heatmap. WinCLIP checks small patches individually at multiple scales.

Zero-Shot Results

91.8% Image AUROC

85.1% Pixel AUROC

Limitations

X Hand-crafted prompts

X No systematic prompt study

X Scores only, no explanations

Takeaway: First zero-shot CLIP approach! But prompts are ad-hoc and it can't explain defects.

AnomalyCLIP — Object-Agnostic Prompt Learning for Zero-Shot AD

Zhou et al. | ICLR 2024 | Learned Prompts

Key Insight & Method

Problem: CLIP focuses on object, not anomaly state

"normal screw" vs "damaged screw" -> CLIP sees "screw"

Solution: Learn object-agnostic prompt embeddings

AnomalyCLIP learns object-agnostic continuous prompt embeddings that capture universal normality vs abnormality, while keeping CLIP frozen.

Results

~94% Image AUROC

Better cross-domain generalization

Limitations

- X Needs training phase for prompt optimization
- X Learned prompts not human-interpretable
- X Still no natural language explanations
- X May overfit to training categories

Takeaway: Learned prompts > hand-crafted, but still no explanations.

AnomalyGPT — Industrial AD Using Large Vision-Language Models

Gu et al. | AAAI 2024 (Oral) | MLLM-Based

Architecture & Method

Four components:

- 1 Frozen Image Encoder (CLIP ViT) -> multi-layer features
- 2 Image Decoder -> patch similarity -> localization
- 3 Prompt Learner -> bridges visual info to LLM
- 4 LLM (Vicuna-7B) -> natural language responses

Training: Synthetic anomalies (NSA) — paste textures on normals

Only decoder + prompt learner + projection trained (efficient!)

Results

94.1% AUROC (1-shot)
+ NL explanations
+ Multi-turn dialogue

Limitations

- X Not truly zero-shot
- X Needs fine-tuning on
synthetic anomaly data
- X GPU required for training

Takeaway: First MLLM for AD — detects + explains! But requires fine-tuning.

MMAD — Multi-Modal Anomaly Detection Benchmark

Jiang et al. | ICLR 2025 | Benchmarking MLLMs

Benchmark Design

- 39,672 questions from 8,366 images
- Sources: MVTec AD, MVTec LOCO, VisA, GoodsAD
- 7 subtasks: classification, localization, type, severity, description, cause, repair

Models Tested

- Proprietary: GPT-4o, Claude, Gemini
- Open-source: LLaVA-NeXT, InternVL2
- Specialized: AnomalyGPT

Key Findings — Validates the Research Gap

- Best model (GPT-4o) only 74.9% accuracy — far below industrial >95% requirement
- Open-source MLLMs perform substantially worse on fine-grained defects
- All models struggle with anomaly-specific tasks vs. object-recognition tasks
- Conclusion: MLLMs alone are NOT sufficient — hybrid approaches needed

Comparative Analysis

Method	Venue	Zero-Shot?	AUROC	Explains?	Key Limitation
PatchCore	CVPR'22	No	99.1%	No	Needs training per category
WinCLIP	CVPR'23	Yes	91.8%	No	Hand-crafted prompts
AnomalyCLIP	ICLR'24	Yes*	~94%	No	Needs prompt training
AnomalyGPT	AAAI'24	No	94.1%	Yes	Requires fine-tuning
MMAD	ICLR'25	---	---	---	Benchmark only

No method achieves ALL of: zero-shot + explanations + open-source + no fine-tuning

Identified Research Gaps

01

No Systematic Prompt Engineering Study

WinCLIP uses ad-hoc prompts, AnomalyCLIP learns prompts — but nobody has studied what makes a good prompt for anomaly detection.

02

Limited Exploration of Open-Source MLLMs

AnomalyGPT needs fine-tuning, GPT-4V is proprietary. Potential of LLaVA with CLIP features is unexplored.

03

No Hybrid VLM + MLLM Pipeline

Current methods use CLIP or MLLMs independently. Combining CLIP's detection with MLLM's explanation has not been explored.

Proposed Research Direction

Two-Stage Hybrid Pipeline: Zero-Shot + Interpretable + Open-Source

Stage 1: VLM Anomaly Scoring

- Pre-trained CLIP (no fine-tuning)
- Systematic prompt engineering:
 - Template variations
 - State word analysis
 - Compositional ensembles
- Multi-scale window features
- Output: anomaly score + heatmap



Stage 2: MLLM Reasoning

- Open-source MLLM (LLaVA)
- Only for flagged anomalous images
- Natural language explanations:
 - Defect type & location
 - Severity assessment
 - Inspection reports
- No fine-tuning required

What makes this novel:

Zero-shot (CLIP) + Explainable (MLLM) + Open-source + No fine-tuning

Thank You

Questions & Feedback Welcome

Prakash Kumar Sarangi | M251250CS | prakash_m251250cs@nitc.ac.in